

17																			
18	X		0		1		2		3		4		5		6		7		8
19	f		20		100		190		280		350		300		220		120		3
20																			
21	X	f	expected frequency $f(x)=Np(x)$			Formula			Rounded expected frequency			Formula							
22		0	20		5.642978112	=B\$34*BINOMDIST(A22,\$B\$35,\$B\$37,0)				6	=ROUND(C22,0)								
23		1	100		46.1822771					46									
24		2	190		165.3561596					165									
25		3	280		338.319759					338									
26		4	350		432.627743					433									
27		5	300		354.0636506					354									
28		6	220		181.1041229					181									
29		7	120		52.93431324					53									
30		8	3		6.768996445					7									
31	Total		1583							1583	=SUM(E22:E30)								
32																			

32			
33			Formula
34	N	1583	=B31
35	n	8	
36	np	4.04548326	=SUMPRODUCT(B22:B30,A22:A30)/B31
37	p	0.505685407	=B36/B35
38	q	0.494314593	=1-B37
39	$P(X<3)$	0.137196093	=BINOMDIST(2,B35,B37,1)
40	$P(X\leq 3)$	0.350916724	=BINOMDIST(3,B35,B37,1)
41	$P(X>3)$	0.649083276	=1-B40
42	$P(X>4)$	0.375787166	=1-BINOMDIST(4,B35,B37,1)
43	$P(4<X<7)$	0.611367983	=BINOMDIST(6,B35,B37,1)-BINOMDIST(3,B35,B37,1)
44			